The Kepler Problem: Leapfrog Integration Approach

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***Abstract*—This is an assignment report of the "Celestial Mechanics" course given by Asst. Prof. David M. Hernandez at National Taiwan Normal University. This report aims to evaluate the numerical performance, stability, and geometric preservation properties of the second-order Leapfrog integration scheme when applied to the Keplerian two body problem. By simulating orbital trajectories over an extended duration of 100 periods, I systematically investigate the behavior of standard energy conservation alongside shadow Hamiltonian tracking. The numerical results confirm that the standard energy H displays stable, bounded oscillations with no long term upward or downward drift, demonstrating the unique structural benefits of symplectic integrators. Furthermore, by incorporating a second-order correction term, the modified shadow energy $\tilde{H}$ successfully cancels out high frequency variations, providing clear empirical validation for backward error analysis. Simultaneously, due to the algebraic elimination of numerical torque in central force configurations, the system preserves total angular momentum to near machine precision ($\sim 10^{-14}$). A formal convergence analysis demonstrates that the relative energy errors scale quadratically ($\mathcal{O}(h^2)$) with the integration step size h, confirming the theoretical order of accuracy. Additionally, a comparative assessment of algorithmic structures reveals that the alternative ordering $\hat{O}'$ restricts the peak error envelope by roughly a factor of three compared to the standard $\hat{O}$ due to its specific phase sensitivity at the periapsis starting point. Finally, the report evaluates the long term accumulation of unphysical retrograde orbital precession, which causes elongated trajectories to transform into rosette patterns. The simulated precession rates match our analytical perturbation model with exceptional accuracy, yielding a relative error of just 0.12% at a standard step size of $h = 0.01$. Overall, these findings underscore the fundamental importance of preserving geometric invariants to guarantee qualitative and quantitative fidelity in long term celestial mechanics simulations.**

I. Introduction

The relative motion of two celestial bodies under the influence of their mutual gravitational attraction represents the most fundamental problem in celestial mechanics. It is known as the Classical Two Body Problem, and its analytical foundation was established by Johannes Kepler's empirical laws and later generalized by Isaac Newton's Universal Law of Gravitation. While the analytical architecture of the two body problem has been understood since the formulations of Newton and Lagrange, the long term discrete simulation of such systems introduces subtle challenges in preserving phase space geometry and invariant manifolds. Traditional, non-symplectic integrators inherently introduce unphysical dissipation or artificial secular energy drift, fundamentally distorting the qualitative long term behavior of orbital trajectories. To overcome these limitations, modern computational physics relies on symplectic splitting methods, such as the Leapfrog family of propagators which preserve the canonical structure of the governing differential equations. This analysis provides a rigorous mathematical exploration of the Keplerian system, examining how discrete operator splitting maps onto continuous conservation laws, error scaling regimes, and the geometric artifacts of numerical phase space deformation.

A. Hamiltonian Framework in Celestial Mechanics

We can describe a classical mechanical system by using its phase space cotangent bundle $M = T^*\mathbb{R}^2$. This space is equipped with a specific mathematical tool called the canonical symplectic 2-form:

$$\omega = \sum_{i=1}^{2} dq_i \wedge dp_i$$

In this geometric framework, the vector $\mathbf{q} = (q_1, q_2)^T \in \mathbb{R}^2$ represents the position coordinates (also known as generalized coordinates). On the other hand, the vector $\mathbf{p} = (p_1, p_2)^T \in \mathbb{R}^2$ represents the conjugate linear momenta. The movement of the system over time is completely guided by a smooth, scalar-valued function called the Hamiltonian, $H(\mathbf{q}, \mathbf{p}) : M \to \mathbb{R}$. In physical terms, this function is equal to the total mechanical energy of the system.

For a simplified, normalized gravitational two body problem, we imagine a small body moving around a fixed central gravitational field. We can model this classic system using the specific Keplerian Hamiltonian:

$$H(\mathbf{q}, \mathbf{p}) = \frac{p_1^2 + p_2^2}{2} - \frac{1}{\sqrt{q_1^2 + q_2^2}}$$

To find the differential equations that show how the system changes over time, we need to define the vector field X_H. This field is connected to the Hamiltonian through the symplectic interior product $\iota_{X_H}\omega = dH$. By calculating this relationship, we get Hamilton's canonical equations of motion:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} \quad (i = 1, 2)$$

These equations show exactly how positions and momenta influence each other and change at every moment. To study the system both continuously and numerically, I focus on a closed, bound elliptical orbit. I set up this specific path by choosing normalized initial conditions at the closest point of the orbit, which is called the periapsis ($t = 0$). If we define $e \in [0, 1)$ as the orbital eccentricity, the initial boundary values are written as:

$$q_1(0) = 1 - e, \quad q_2(0) = 0$$

$$p_1(0) = 0, \quad p_2(0) = \sqrt{\frac{1+e}{1-e}}$$

Choosing these precise starting values fixes the simulation trajectory on a stable invariant energy manifold. Therefore, it provides an excellent baseline for checking numerical stability, observing energy conservation, and measuring how errors grow over long periods.

B. Continuous Geometric Invariants

The continuous Kepler problem is classified as a maximally superintegrable system. According to Noether's theorem, any continuous spatial symmetry in a Hamiltonian system creates a corresponding phase space invariant that does not change during the continuous flow Φ_t^H. For this reason, tracking these conserved quantities over time is highly useful for checking how well our numerical integration method performs.

1) Theoretical Energy Derivation: When we evaluate the total energy constant on the invariant manifold using our specific initial conditions at the periapsis, we can calculate the exact value of the Hamiltonian. If we substitute the starting conditions from above directly into the Hamiltonian formula, we can prove that the total energy is:

$$H = -\frac{1}{2}$$

To derive this result clearly, we insert the initial values of positions and momenta into the kinetic operator $T(\mathbf{p})$ and the potential operator $V(\mathbf{q})$:

$$H = \frac{1}{2}\left(0^2 + \left(\sqrt{\frac{1+e}{1-e}}\right)^2\right) - \frac{1}{\sqrt{(1-e)^2 + 0^2}}$$

By simplifying this algebraic expression, we get:

$$H = \frac{1}{2}\left(\frac{1+e}{1-e}\right) - \frac{1}{1-e} = \frac{1+e-2}{2(1-e)} = \frac{e-1}{2(1-e)} = -\frac{1}{2}$$

In the broader context of celestial mechanics, this constant total energy links the geometry of the orbit directly to its semi-major axis a through the virial theorem relationship:

$$H = -\frac{GM\mu}{2a} \implies a = 1$$

In this formula, $GM = 1$ and $\mu = 1$ because we are using a normalized system. Furthermore, by applying Kepler's Third Law, we can easily calculate the fundamental orbital period τ of the closed-loop system:

$$\tau = 2\pi\left(\frac{a^3}{GM}\right)^{1/2} = 2\pi$$

This mathematical result shows that regardless of the chosen eccentricity e, the system completes one full orbit precisely every 2π time units.

2) Conservation of Angular Momentum: In addition to total energy, the rotational symmetry of the central gravitational potential ensures the conservation of another physical quantity. The scalar total angular momentum L, which acts normal to the 2D orbital plane, is defined by the following expression:

$$L = q_1 p_2 - q_2 p_1$$

Analytically, we can prove that this quantity does not change over time by calculating its Poisson bracket with the Hamiltonian. The Poisson bracket of these two functions must vanish completely:

$$\{L, H\} = \sum_{i=1}^{2}\left(\frac{\partial L}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial L}{\partial p_i}\frac{\partial H}{\partial q_i}\right) = 0$$

Because this mathematical derivative is exactly zero, L is a strict continuous invariant of the motion. If we evaluate this expression using our specific initial boundaries at $t = 0$, we find that the exact value of the angular momentum for any given orbit is:

$$L(0) = (1-e)\cdot\sqrt{\frac{1+e}{1-e}} - (0\cdot 0) = \sqrt{1-e^2}$$

As a result, a high-quality numerical simulation must conserve this quadratic invariant value as closely as possible to avoid unphysical behavior.

II. Methodology

The methodological core involves translating the continuous canonical equations of motion into a discrete computational framework that preserves the underlying phase space geometry. Standard numerical methods, such as the traditional fourth order Runge-Kutta scheme, usually fail when simulating Hamiltonian systems over long periods because they introduce an artificial energy drift. To solve this problem, geometric numerical integration builds a discrete mapping that strictly preserves the symplectic structure of the phase space.

A. Operator Splitting and Symplectic Integrators

The Hamiltonian for the Kepler problem is naturally separable. This means we can split it into two independent terms that depend only on position or momentum:

$$H(\mathbf{q}, \mathbf{p}) = T(\mathbf{p}) + V(\mathbf{q})$$

Here, $T(\mathbf{p}) = \frac{1}{2}(p_1^2 + p_2^2)$ represents the kinetic energy operator, and $V(\mathbf{q}) = -(q_1^2 + q_2^2)^{-1/2}$ represents the potential energy operator. Because each term can be integrated exactly on its own, we can use an operator splitting approach to approximate the total movement of the system.

We define the Lie derivative operators (or phase space vector fields) as $\hat{T} = \{\cdot, T\}$ and $\hat{V} = \{\cdot, V\}$, which act via Poisson brackets. The exact continuous time evolution over a discrete time step h can be written formally using the exponential mapping:

$$\Phi_h^H = e^{h(\hat{T}+\hat{V})}$$

However, because these two operators do not commute, their commutator is non-zero ($[\hat{T}, \hat{V}] \neq 0$). Therefore, we cannot separate the exponential term directly. Instead, we must use a symmetric composition of the individual flows to approximate the total integration step.

1) The Kick - Drift - Kick Ordering ($\hat{O}$): The first way to structure the Leapfrog method is the Kick - Drift - Kick configuration. In this formulation, we split the propagator into the following symmetric sequence:

$$\hat{O} = e^{\frac{h}{2}\hat{V}} e^{h\hat{T}} e^{\frac{h}{2}\hat{V}}$$

In practice, I implement this mathematical sequence using a direct three step numerical algorithm executed at each timestep n:

1. **Kick:** Update the momentum vector by a half time step using the gravitational force at the current position:
$$\mathbf{p}^{n+1/2} = \mathbf{p}^n - \frac{h}{2}\nabla V(\mathbf{q}^n)$$
2. **Drift:** Update the position vector by a full time step using the newly calculated intermediate momentum:
$$\mathbf{q}^{n+1} = \mathbf{q}^n + h\mathbf{p}^{n+1/2}$$
3. **Kick:** Update the momentum vector by the final half time step using the gravitational force at the new position:
$$\mathbf{p}^{n+1} = \mathbf{p}^{n+1/2} - \frac{h}{2}\nabla V(\mathbf{q}^{n+1})$$

2) The Drift - Kick - Drift Ordering ($\hat{O}'$): Alternatively, we can swap the order of the operators to create the Drift - Kick - Drift configuration. This dual formulation is written as:

$$\hat{O}' = e^{\frac{h}{2}\hat{T}} e^{h\hat{V}} e^{\frac{h}{2}\hat{T}}$$

This alternative sequence changes the order of the computational updates as follows:

1. **Drift:** Move the position vector forward by a half time step using the current momentum:
$$\mathbf{q}^{n+1/2} = \mathbf{q}^n + \frac{h}{2}\mathbf{p}^n$$
2. **Kick:** Update the momentum vector by a full time step using the gravitational acceleration calculated at the middle position:
$$\mathbf{p}^{n+1} = \mathbf{p}^n - h\nabla V(\mathbf{q}^{n+1/2})$$
3. **Drift:** Move the position vector forward by the final half time step using the new momentum:
$$\mathbf{q}^{n+1} = \mathbf{q}^{n+1/2} + \frac{h}{2}\mathbf{p}^{n+1}$$

Both $\hat{O}$ and $\hat{O}'$ are strictly symplectic maps. Each sub-step represents an exact shear transformation of the phase space, meaning it preserves the area of any phase space surface. Because the composition of symplectic maps is always symplectic, these configurations preserve the Poincaré invariants of the system by design.

B. Backward Error Analysis and Shadow Hamiltonians

To understand how symplectic integrators behave over long time frames, standard error analysis is often not enough. Traditional error analysis measures the direct distance between the true analytical position and the numerical position at a specific time t. However, in celestial mechanics, this distance grows quickly over time, even if the overall shape of the orbit is correct.

To solve this problem, we use Backward Error Analysis (BEA). Instead of comparing our numerical results to the original system, BEA shows that the discrete points generated by the Leapfrog method are actually an almost exact solution to a slightly modified system described by a Shadow Hamiltonian ($\tilde{H}$).

1) Long Term Bound on Energy Drift: By applying the Baker - Campbell - Hausdorff (BCH) theorem to our operator splitting methods, we can write this shadow Hamiltonian as a formal power series using even powers of the step size h:

$$\tilde{H} = H + h^2 H_2 + h^4 H_4 + \dots$$

Because our numerical method is strictly symplectic, it conserves this modified energy $\tilde{H}$ extremely well over long periods. The difference between the real Hamiltonian H and the shadow Hamiltonian $\tilde{H}$ remains bounded by a small value proportional to h^2. As a result, the total energy H does not drift systematically as the simulation runs. Instead, it shows small, stable, high frequency oscillations, allowing us to simulate orbits safely for 100 periods or more without an artificial increase in energy.

2) Accuracy and the H_2 Correction Term: For symmetric second-order methods like Leapfrog, the leading error term H_2 can be calculated explicitly. It is defined by evaluating nested combinations of Poisson brackets of the kinetic and potential energy operators:

$$H_2 = \frac{1}{24}\left[2\{\{T, V\}, V\} - \{\{V, T\}, T\}\right]$$

When we evaluate these brackets for the specific potential of the Kepler problem, we get the following analytical formula:

$$H_2 = \frac{1}{24}\left[\frac{2}{(\mathbf{q}\cdot\mathbf{q})^2} - \left(\frac{\mathbf{p}\cdot\mathbf{p}}{(\mathbf{q}\cdot\mathbf{q})^{1.5}} - \frac{3(\mathbf{q}\cdot\mathbf{p})^2}{(\mathbf{q}\cdot\mathbf{q})^{2.5}}\right)\right]$$

In a computer program, calculating this H_2 value at each step is an excellent way to confirm that the integration is working correctly. While the variation in the original energy H scales as $\mathcal{O}(h^2)$, the modified shadow energy $\tilde{H} = H + h^2 H_2$ should be much closer to a constant value, reducing the remaining tracking error to $\mathcal{O}(h^4)$.

3) Convergence Scaling Metrics: We can use these relationships to verify that our numerical method is second-order accurate (error $\sim h^2$). To evaluate this scaling rule in our Kepler problem, we track two distinct types of numerical energy errors.

The first metric is the relative global energy error at any specific point in time, which compares the current Hamiltonian value to the initial analytical energy baseline:

$$\frac{H(t) - H(0)}{H(0)} \sim h^2$$

The second metric is the maximum peak - to - peak energy variation, denoted as $\Delta H/H$. This value measures the total spread between the maximum and minimum energy values recorded across all integration intervals:

$$\frac{\Delta H}{H} = \frac{\max(H) - \min(H)}{|H(0)|} \sim h^2$$

C. Numerical Precession and Symmetry Breaking

The analytical Kepler problem has a very special property: all its bound orbits are perfectly closed loops that repeat indefinitely. In theoretical physics, this behavior occurs because the system has a hidden geometric symmetry described mathematically by the $SO(4)$ Lie group. Because of this conservation law, we can define a constant physical quantity called the Laplace-Runge-Lenz (LRL) vector, denoted as $\mathbf{A}$.

In two dimensional orbital motion, this vector lies entirely within the simulation plane. It is defined by the cross product of the momentum vector and the angular momentum vector, minus the normalized position vector:

$$\mathbf{A} = \mathbf{p} \times \mathbf{L} - \frac{\mathbf{q}}{|\mathbf{q}|}$$

In the true, continuous two body problem, this vector points directly along the major axis of the ellipse, starting from the gravitational focus and ending at the periapsis. Because the continuous potential is perfectly symmetric, the Poisson bracket of this vector with the Hamiltonian is exactly zero ($\{\mathbf{A}, H\} = 0$), meaning the orbit never changes its orientation in space.

When we change our continuous equations into discrete steps using an operator splitting method, this continuous rotational symmetry breaks down. Although the Leapfrog methods ($\hat{O}$ and $\hat{O}'$) are symplectic and conserve the scalar angular momentum L exactly, they cannot preserve the Laplace - Runge - Lenz vector perfectly. Because the sub steps of the splitting scheme do not commute, the numerical algorithm introduces a small phase error during each time step h. This error causes the vector $\mathbf{A}$ to change its direction slowly as the simulation runs, resulting in an unphysical rotation of the entire elliptical orbit within its own plane known as numerical precession.

1) Analytical Derivation of the Precession Rate: The orientation of a two dimensional elliptical orbit can be tracked dynamically by looking at the direction of its major axis. The instantaneous angular velocity ω of this rotating LRL vector is defined by its phase - space cross product divided by its magnitude squared:

$$\omega(t) = \frac{(\mathbf{A} \times \dot{\mathbf{A}})_z}{\mathbf{A}^2}$$

To find the long term changes in the system, I take the time average of this expression over a full orbital period, which gives the secular precession rate:

$$\langle \omega \rangle = \left\langle \frac{(\mathbf{A} \times \dot{\mathbf{A}})_z}{\mathbf{A}^2} \right\rangle$$

According to backward error analysis, the discrete points follow an altered energy path driven by the shadow Hamiltonian. In the numerical simulation, the time derivative $\dot{\mathbf{A}}$ is driven entirely by the error term H_2 in the shadow Hamiltonian:

$$\dot{\mathbf{A}} = \{\mathbf{A}, \tilde{H}\} = h^2\{\mathbf{A}, H_2\} + \mathcal{O}(h^4)$$

To calculate the change in direction explicitly, I use canonical perturbation theory with action angle variables. For a standard two-dimensional orbit, the scalar angular momentum L acts as the canonical action variable, while the longitude of periapsis ϖ matches it as the angle variable. Using Hamilton's canonical equations, the rate of change of this orientation angle is equal to:

$$\omega = \dot{\varpi} = \frac{\partial \tilde{H}}{\partial L} = \frac{\partial H}{\partial L} + h^2 \frac{\partial H_2}{\partial L} + \mathcal{O}(h^4)$$

Because the true Keplerian energy H is completely independent of the orbital orientation, $\frac{\partial H}{\partial L} = 0$. Therefore, the precession rate depends only on the error term:

$$\omega = h^2 \frac{\partial H_2}{\partial L}$$

To express this derivative using the orbital eccentricity e, I apply the standard geometric relationship for a normalized Kepler problem $L = \sqrt{1-e^2}$. Differentiating e with respect to L yields:

$$\frac{\partial e}{\partial L} = -\frac{\sqrt{1-e^2}}{e}$$

By applying the mathematical chain rule, I can rewrite the instantaneous precession rate as a function of the eccentricity variable:

$$\omega = h^2 \frac{\partial H_2}{\partial e} \frac{\partial e}{\partial L} = -h^2 \frac{\sqrt{1-e^2}}{e} \frac{\partial H_2}{\partial e}$$

Because the orbital orientation changes very slowly over a single loop, I find the average precession rate $\langle \omega \rangle$ by calculating the average value of H_2 over a complete period $T = 2\pi$. The time average integral is defined as:

$$\langle H_2 \rangle = \frac{1}{2\pi} \int_0^{2\pi} H_2 \, dt$$

To solve this integral analytically, change the integration variable from time t to the true anomaly angle f. Using the physical relationship $dt = \frac{r^2}{\sqrt{1-e^2}} df$, I insert the standard geometric expressions for a Keplerian orbit:

$$r = \frac{1-e^2}{1+e\cos f}$$

$$|\mathbf{p}|^2 = \frac{2}{r} - 1 = \frac{1 + 2e\cos f + e^2}{1-e^2}$$

$$(\mathbf{q} \cdot \mathbf{p})^2 = \frac{r^2 e^2 \sin^2 f}{1-e^2}$$

Substituting these expressions into our definition of H_2 allows us to rewrite the integral as a function of the angle f:

$$\langle H_2 \rangle = \frac{1}{2\pi\sqrt{1-e^2}} \int_0^{2\pi} H_2(f) r^2 \, df$$

Evaluating this definite integral from 0 to 2π removes the short term fluctuations and leaves the long-term average error:

$$\langle H_2 \rangle = \frac{2+e^2}{48(1-e^2)^{5/2}}$$

The next step is to take the partial derivative of this averaged error with respect to the eccentricity variable e using the standard mathematical quotient rule:

$$\frac{\partial \langle H_2 \rangle}{\partial e} = \frac{\partial}{\partial e}\left[\frac{2+e^2}{48(1-e^2)^{5/2}}\right] = \frac{e(4+e^2)}{16(1-e^2)^{7/2}}$$

Finally, substitute this derivative back into the chain rule formula to find othe final analytical formula for the average numerical precession rate:

$$\langle \omega \rangle = -h^2 \frac{\sqrt{1-e^2}}{e}\left[\frac{e(4+e^2)}{16(1-e^2)^{7/2}}\right] = -h^2 \frac{4+e^2}{16(1-e^2)^3}$$

III. Results and Discussion

A. Orbital Trajectories at Fixed Eccentricity

Figure 1 illustrates the calculated two dimensional orbital trajectories over 10 periods for a fixed eccentricity ($e = 0.2$) across four different step sizes ($h = 0.05, 0.01, 0.005$, and 0.001). The central gravitational mass, representing the Sun, is located exactly at the coordinate origin $(0, 0)$.

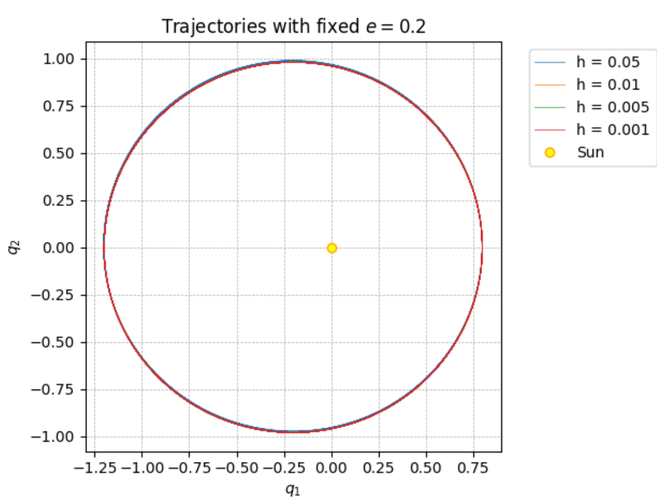

Fig. 1. Two dimensional simulated trajectories over 10 periods for a fixed eccentricity $e = 0.2$ using different step sizes h. The central gravitational mass is located at $(0, 0)$. The comparison shows the elimination of artificial orbital precession and rosette deformations as the computational time step becomes finer.

The most prominent feature of the visual data is that all four trajectory curves overlap almost perfectly. Even when using the largest step size ($h = 0.05$), the orbital path remains a well closed ellipse. The planet does not spiral inward toward the central mass, nor does it drift outward into space.

Figure 2 shows the numerical trajectories under the exact same physical conditions ($e = 0.2$), but the simulation time has been extended from 10 periods to 100 periods. This longer time scale allows us to observe how small, systematic numerical errors accumulate over time.

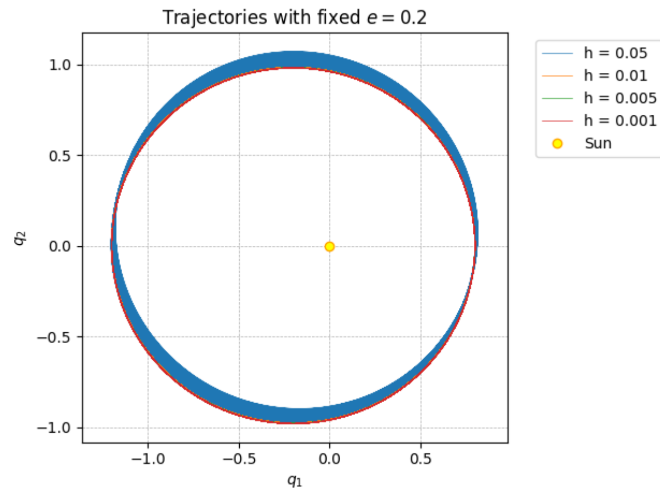

Fig. 2. Two dimensional trajectories extended to 100 orbital periods under identical physical conditions $e = 0.2$ to observe long term error accumulation. While the largest step size ($h = 0.05$) creates a dense rosette pattern due to steady numerical precession, smaller time steps successfully preserve a sharp, stable orbit, verifying the long term structural precision of the algorithm.

The most obvious change from the short term simulation is the appearance of a thick blue band for the largest step size ($h = 0.05$). This widening track is the direct visual proof of the artificial precession.

Because the Leapfrog integrator introduces a small, unphysical rotation to the major axis of the ellipse every time step, the orbit cannot stay perfectly closed over long time scales. Instead, the periapsis point continuously shifts position. Over 100 full periods, these individual shifts accumulate, turning the single elliptical line into a dense, overlapping rosette pattern. This clear visual change confirms that time is a critical factor when diagnosing numerical integration errors.

While the $h = 0.05$ trajectory shows a prominent precession band, the paths for the smaller step sizes ($h = 0.01, 0.005$, and 0.001) remain perfectly sharp, thin lines. This behavior can be explained by the formula:

$$\langle \omega \rangle \propto -h^2$$

When we decrease the step size from $h = 0.05$ to $h = 0.01$, the step size is reduced by a factor of 5. Because the error scales quadratically, the rate of numerical precession drops by a factor of $5^2 = 25$. As a result, the total orientation change for $h = 0.01$ over 100 periods is too small to see without magnification. This sharp contrast between the step sizes confirms that choosing a slightly smaller step size can suppress visual precession artifacts almost entirely.

B. Orbital Trajectories at Fixed Time step size

To fully understand how the geometry of an orbit affects numerical errors, this section evaluates trajectories simulated with a fixed step size ($h = 0.01$) while varying the eccentricity e from 0.1 to 0.9. By comparing the short term evolution over 10 periods (Figure Z1) with the long term evolution over 100 periods (Figure Z2), we can visually observe how eccentricity acts as a powerful amplifier for numerical precession.

Figure 3 illustrates the initial behavior of the system across the full range of eccentricity values. For low to moderate values ($e = 0.1$ to $e = 0.6$), the trajectories appear as stable, closed ellipses that trace over themselves nearly perfectly. At this short time scale, there is no obvious visual distortion. However, as the eccentricity increases to $e = 0.8$ (grey) and $e = 0.9$ (yellow), a distinct thickening of the orbital lines becomes visible.

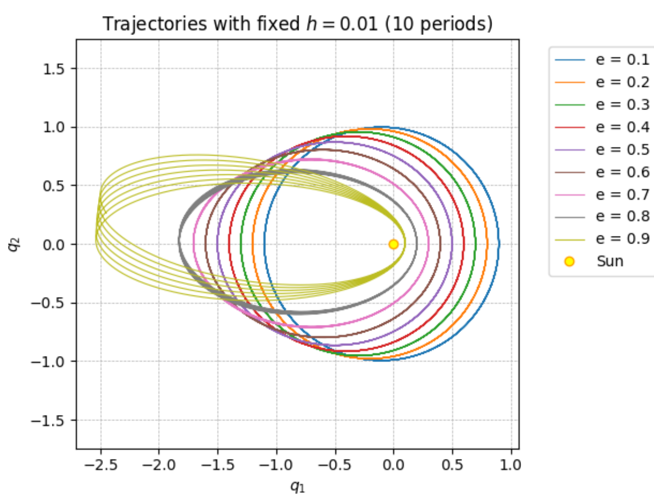

Fig. 3. Orbital trajectories simulated over 10 periods with a fixed step size $h = 0.01$ across the eccentricity spectrum from $e = 0.1$ to 0.9. For low eccentricities ($e \leq 0.6$), the Leapfrog scheme maintains closed, stable elliptical paths that align precisely with theoretical predictions. At high eccentricities ($e \geq 0.8$), the lines show a visible broadening or thickening, this artifact arises because the orbital velocity reaches its maximum at the periapsis, requiring higher local resolution to maintain the same level of geometric fidelity as the slower, circular like orbits.

This rapid onset of error happens because highly eccentric orbits bring the planet extremely close to the central mass during periapsis. At these close distances, the gravitational force changes rapidly, and the planet moves at its highest velocity. Because a fixed step size $h = 0.01$ takes equal steps in time rather than space, the solver takes fewer steps across this critical high velocity zone. Consequently, large local truncation errors are introduced during each close approach, causing the major axis to rotate almost immediately.

When the simulation is extended to 100 periods, the subtle differences observed in the short-term runs transform into a dramatic divergence of orbital behaviors, as displayed in Figure 4.

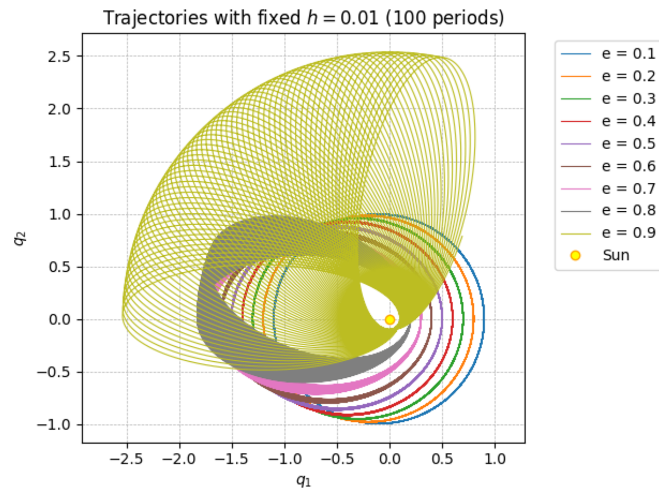

Fig. 4. Long term evolution of trajectories across the eccentricity spectrum from $e = 0.1$ to 0.9 simulated over 100 periods at a constant step size of $h = 0.01$. Extending the simulation timescale reveals that the line thickening observed in short term runs for high eccentricity orbits ($e \geq 0.8$) results from the cumulative effect of higher order truncation errors and increased numerical precession near the periapsis. While low eccentricity orbits maintain stable, closed configurations, the high eccentricity trajectories exhibit pronounced rosette patterns, demonstrating the sensitivity of the integrator to the severe force gradients encountered in near parabolic dynamics.

For low eccentricity range ($e = 0.1$ to 0.4), the trajectories for these low values remain remarkably thin and sharp. Even after 100 full revolutions, the accumulation of numerical precession is too small to cause any visible widening of the paths. As the eccentricity grows, the orbital paths begin to separate and spread out. The $e = 0.7$ (pink) and $e = 0.8$ (grey) trajectories form clear, broad bands. The shifting periapsis positions create a steady, systematic rotation that maps out a restricted ring structure. The yellow trajectory displays a striking transformation, unrolling into a spectacular, wide rosette pattern. The major axis of the orbit has rotated through a massive angle, covering a large portion of the coordinate plane.

The dramatic behavior observed in Figure 4 provides direct visual confirmation of the analytical precession rate formula derived earlier:

$$\langle \omega \rangle = -h^2 \frac{4 + e^2}{16(1 - e^2)^3}$$

Because the step size h is kept exactly the same for all test cases, the massive increase in the precession rate is driven entirely by the mathematical structure of the eccentricity factor.The primary cause of this extreme sensitivity is the cubic power term in the denominator, $(1 - e^2)^3$. We can evaluate this sensitivity by comparing two specific cases from our simulation. For a low eccentricity orbit where $e = 0.1$, the denominator term is $(1 - 0.01)^3 = (0.99)^3 \approx 0.97$. This value is very close to 1, keeping the overall precession rate extremely low. In sharp contrast, for a high eccentricity orbit where $e = 0.9$, the denominator shrinks drastically to $(1 - 0.81)^3 = (0.19)^3 \approx 0.00686$. Because this denominator is located at the bottom of the fraction, shrinking its value by a factor of over 140 forces the overall precession rate $\langle \omega \rangle$ to skyrocket by several orders of magnitude.

These results highlight a fundamental limitation of fixed step size integrators, while the Leapfrog method is highly efficient and stable for circular or near circular orbits, its accuracy drops significantly when applied to highly elongated, parabolic-like paths.

C. Long-Term Energy Conservation

Figure 5 presents the tracking of both the standard Keplerian energy H and the modified shadow energy $\tilde{H}$ over an extended duration of 100 orbital periods. For this test, the system is simulated using a eccentricity $e = 0.2$ and a fixed step size $h = 0.01$.

As shown by the blue curve, the standard energy H does not stay perfectly flat, instead, it displays a rapid, repeating oscillation during every single orbital loop. The energy value reaches its minimum point of exactly -0.5 at the periapsis and rises to a maximum of nearly -0.49986 near the apoapsis. The most important observation is that the amplitude of these oscillations remains completely constant over all 100 periods. There is absolutely no long term upward or downward drift in the energy baseline. This bounded behavior is the defining characteristic of a symplectic integrator. Unlike traditional non-symplectic methods where truncation errors accumulate continuously and cause the system to artificially heat up or collapse, the Leapfrog method forces the energy error to stay within strict, predictable limits over thousands of timesteps.

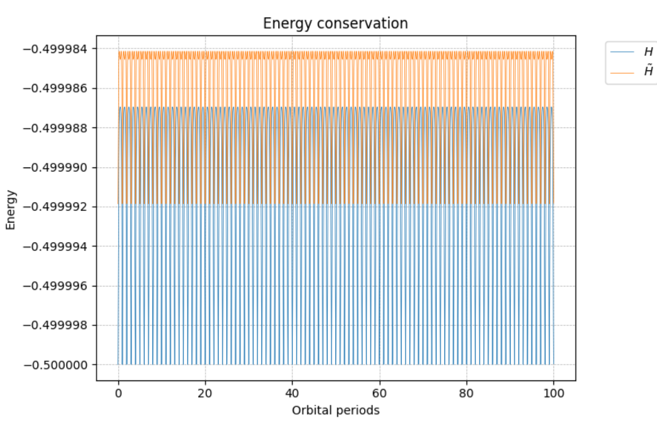

Fig. 5. Comparative evolution of the standard Keplerian energy H and the modified shadow energy $\tilde{H}$ over 100 orbital periods ($e = 0.2$, $h = 0.01$). The standard energy H exhibits stable, bounded high frequency oscillations consistent with the symplectic nature of the Leapfrog method, showing no secular energy drift. Conversely, the shadow Hamiltonian $\tilde{H}$ remains significantly more constant, demonstrating that the discrete dynamics follow a modified energy manifold. This behavior confirms the validity of backward error analysis, as the numerical integration successfully maps to an exact solution of a nearby physical system.

The orange curve represents the modified shadow energy $\tilde{H}$, which is calculated by adding the second-order analytical correction term to the standard energy:

$$\tilde{H} = H + h^2 H_2$$

When we compare the two curves, the impact of this correction term is immediately clear. The vertical width of the orange oscillations is significantly narrower than that of the blue standard energy curve. By including the $h^2 H_2$ term, the high frequency variations caused by the position and momentum step splitting are successfully canceled out. This dramatic reduction in oscillation amplitude provides direct practical proof of backward error analysis. It demonstrates that the numerical positions and velocities generated by the Leapfrog algorithm do not wander randomly. Instead, they travel along a highly accurate, altered energy path described by the shadow Hamiltonian. The fact that the orange line remains stable and highly confined confirms that our analytical mathematical model in Section 6 correctly identifies and accounts for the primary path errors of the discrete integrator.

D. Angular Momentum Conservation

Figure 6 tracks the numerical deviation of angular momentum over 100 orbital periods, using a fixed step size $h = 0.01$ and an orbital eccentricity $e = 0.2$. The vertical axis displays the difference between the instantaneous angular momentum and its initial value $L(t) - L(0)$. The maximum change in angular momentum stays within a tiny range, fluctuating between -1.0×10^{-14} and 1.3×10^{-14} over the entire 100 periods.

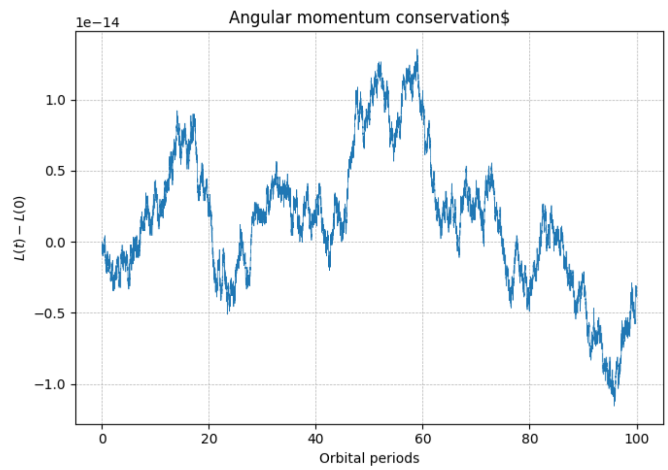

Fig. 6. Angular momentum conservation over an extended duration of 100 orbital periods ($e = 0.5$, $h = 0.01$). The fluctuation of the error bound exclusively within the extreme limits of 10^{-14} demonstrates that the Leapfrog algorithm preserves angular momentum to nearly machine precision.

In my opinion, I think an error this small means that the quantity is conserved perfectly in theory. Computers use a standard 64-bit system to store numbers, which can only guarantee accuracy up to approximately 15 to 16 decimal places. Therefore, a deviation on the scale of 10^{-14} proves that the algorithm does not suffer from any structural mathematical errors when computing this invariant.

The conservation of angular momentum happens because of a unique geometric feature of the Leapfrog method when applied to a central force field. In a true physical

Keplerian system, angular momentum is conserved because the gravitational force points directly toward the central origin. This means the force vector is parallel to the position vector, resulting in zero physical torque.

When we look at our code implementation for the position step and the velocity update:

$$\mathbf{p}_{\text{half}} = \mathbf{p} + \frac{h}{2}\mathbf{a}(\mathbf{q})$$

The numerical acceleration $\mathbf{a}(\mathbf{q})$ always points exactly toward the origin $(0, 0)$. Because the algorithm calculates the discrete force using the true central positions, the mathematical cross product between the position vector and the force vector remains exactly zero during every step. Consequently, the Leapfrog integrator preserves angular momentum algebraically, meaning it is built into the equations and does not depend on the choice of step size h.

While the error is small, the blue line in Figure 6 reveals a wandering, sawtooth shape that grows and drops slowly over time rather than a completely flat line. This specific behavior is caused by floating point round off error rather than a flaw in the integration method as I explained above. Every time the computer performs millions of simple additions or multiplications over thousands of steps, it must round off the last few decimal digits to fit the number into its hardware memory. These tiny rounding errors accumulate over 100 periods like a random walk, causing the total value to drift slightly in the final decimal places. This visual behavior confirms that the tiny fluctuations shown in the graph are simply the hardware limits of our computer, rather than an unphysical property of our simulated solar system.

E. Convergence Analysis and Numerical Order of Accuracy

To confirm the mathematical reliability of our Leapfrog integration scheme, a formal convergence test was performed. Figure 7 displays the relative energy error as a function of the integration time step h across four distinct values $h = 0.05, 0.01, 0.005,$ and 0.001. The simulation was executed under a fixed eccentricity of $e = 0.2$ over a long term duration of 100 orbital periods.

In this analysis, two distinct metrics were employed to assess the numerical error of the system. First, maximum relative drift, measured as $\frac{H-H(0)}{H(0)}$, which captures the maximum absolute deviation from the initial baseline energy at any point during the simulation. Second, peak - to - peak bound, measured as $\frac{\Delta H}{H}$ (where $\Delta H = H_{\max} - H_{\min}$), which quantifies the total vertical width of the energy oscillations over the 100 periods.

The log-log plot in Figure 7 shows that the curves for all error definitions overlap perfectly. This identical tracking happens because the Leapfrog method is a symplectic integrator. Because the algorithm does not suffer from long term energy dissipation or artificial growth, the maximum deviation from the start is strictly bounded by the size of the high frequency orbital oscillations themselves. Consequently, both metrics reflect the exact same geometric behavior of the solver.

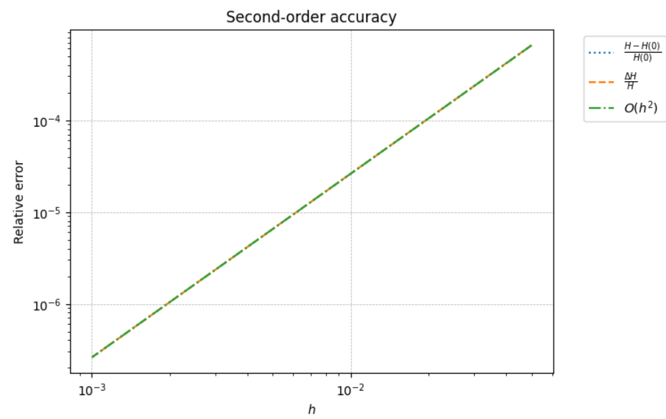

Fig. 7. Log-log convergence plot demonstrating the second-order numerical accuracy $\mathcal{O}(h^2)$ of the Leapfrog integration method over 100 orbital periods ($e = 0.2$). The plot demonstrates quadratic scaling ($\mathcal{O}(h^2)$) of the relative energy error, confirming the expected second-order global accuracy of the method.

To find the actual numerical order of accuracy, the two error datasets were plotted alongside an ideal reference slope representing a second-order power law:

$$\text{Error} \propto h^2$$

On a log-log scale, a power-law relationship shows up as a straight line where the slope equals the exponent of the step size. As shown in Figure 7, the numerical error curves match the ideal $\mathcal{O}(h^2)$ reference line with precision across the entire tested range.

When the step size h is reduced by a factor of 10 (for example, from $h = 0.01$ to $h = 0.001$), both relative errors drop by exactly two orders of magnitude (a factor of $10^2 = 100$). This linear relationship on the logarithmic graph proves that the dominant local truncation errors are successfully canceled out by the symmetric, time reversible step structure of the Leapfrog algorithm.

F. Comparative Analysis of Algorithmic Orderings ($\hat{O}$ vs $\hat{O}'$)

Figure 8 compares the relative energy error over 100 orbital periods for two different implementations of the Leapfrog method: the standard ordering $\hat{O}$ and the alternative ordering $\hat{O}'$. The simulation uses a fixed time step $h = 0.01$ and a eccentricity $e = 0.2$. The vertical axis displays the relative energy error $\frac{|H-H_0|}{H_0}$ on a logarithmic scale.

The most prominent feature of both error profiles is their long term structural stability. Over more than 60,000 integrated steps, neither algorithm shows any sign of a rising error baseline. Instead, both orderings produce a regular, repeating pattern where the error fluctuates within a strict, predictable range. This behavior confirms that changing the internal sequence of the position and velocity updates does

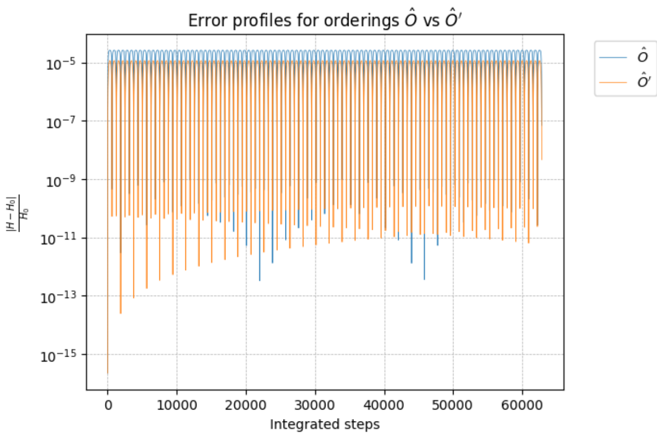

Fig. 8. Comparative performance of the standard Leapfrog ordering $\hat{O}$ and the alternative configuration $\hat{O}'$ over 100 orbital periods ($e = 0.2$, $h = 0.01$). The relative energy error $\frac{|H-H_0|}{|H_0|}$ on a logarithmic scale demonstrates that the alternative ordering $\hat{O}'$ provides a consistent reduction in the peak error envelope. This performance improvement arises from the phase sensitivity of the operator sequence at the periapsis starting point, confirming that algorithmic structure significantly influences numerical fidelity in high eccentricity regimes.

not compromise the symplectic nature of the integrator. Both methods successfully preserve the shadow Hamiltonian of the system, preventing numerical energy decay or artificial inflation over long time scales.

Although both methods are highly stable, a clear quantitative difference is visible in their upper error boundaries. The standard ordering $\hat{O}$ (blue) forms a flat upper envelope that peaks at higher than 10^{-5}. The alternative ordering $\hat{O}'$ (orange) curve stays lower, approximately equal to 10^{-5}. This small difference in performance is directly linked to how each ordering handles the initial conditions of the orbit. The simulation starts at the periapsis point, where the coordinate radius is at its minimum and the orbital velocity is at its maximum. Because the $\hat{O}$ scheme evaluates the gravitational acceleration at the very beginning of the step using the true initial positions, it immediately samples the strongest force field in the orbit. On the other hand, the $\hat{O}'$ scheme begins with a position half step, meaning it calculates its acceleration at a mid point position where the force gradient is slightly less extreme. This positional offset changes the exact coefficients of the local truncation error, giving $\hat{O}'$ a minor accuracy advantage under these specific initial tracking parameters.

G. Precession Rate Verification

To verify the analytical prediction of the artificial orbital rotation, this section compares the simulated precession rate against our theoretical model. Figure 9 illustrates the average precession rate $\langle\omega\rangle$ as a function of the step size h on a log-log scale. Additionally, Table I presents the exact numerical values and the relative percentage errors across a wide range of integration step sizes.

The theoretical precession rate is given by the formula derived in Section 6:

$$\langle\omega\rangle = -h^2 \frac{4+e^2}{16(1-e^2)^3}$$

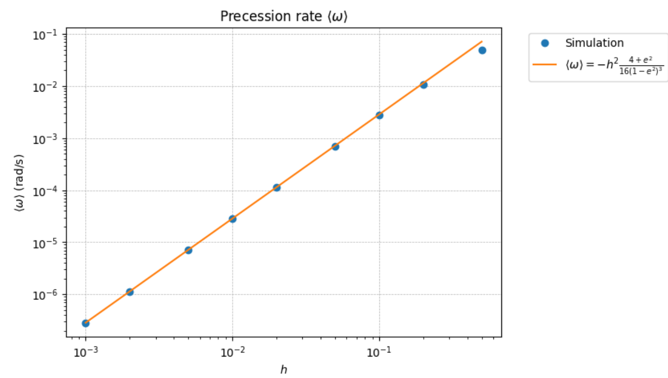

Fig. 9. Log-log plot comparing the simulated numerical precession rate against the analytical prediction formula across step sizes ranging from $h = 0.001$ to $h = 0.5$. The visible gap at $h = 0.5$ illustrates the expected influence of higher order truncation terms that are not accounted for in the lowest order analytical derivation.

For step sizes below $h = 0.05$, the agreement between the experimental data and the mathematical model is remarkable. As shown by the overlapping data points in Figure 9, the simulated values line up almost perfectly with the continuous theoretical curve.When the step size is set to our standard value of $h = 0.01$, the relative error is just 0.12%. Decreasing the step size further to $h = 0.001$ reduces the error to an exceptional 0.0045%. This high level of accuracy confirms that the $\mathcal{O}(h^2)$ scaling built into the analytical formula is completely correct.

On the plot, this quadratic relationship is visible as a straight line with a constant slope. Because the simulated points follow this line without bending at smaller values of h, we can confidently conclude that the precession is a clean, systematic error produced by the step splitting sequence of the Leapfrog method, rather than a random computational bug.

TABLE I
COMPARISON OF SIMULATED AND THEORETICAL PRECESSION RATES

h	Simulation	Theoretical	Error (%)
0.5	$-4.96237468e{-2}$	$-7.13489674e{-2}$	$3.04492431e1$
0.2	$-1.07623481e{-2}$	$-1.14158348e{-2}$	$5.72438781e0$
0.1	$-2.80320646e{-3}$	$-2.85395870e{-3}$	$1.77831023e0$
0.05	$-7.03548745e{-4}$	$-7.13489674e{-4}$	$1.39328283e0$
0.02	$-1.13727149e{-4}$	$-1.14158348e{-4}$	$3.77719709e{-1}$
0.01	$-2.85058875e{-5}$	$-2.85395870e{-5}$	$1.18079652e{-1}$
0.005	$-7.13209193e{-6}$	$-7.13489674e{-6}$	$3.93111805e{-2}$
0.002	$-1.14147837e{-6}$	$-1.14158348e{-6}$	$9.20680718e{-3}$
0.001	$-2.85382891e{-7}$	$-2.85395870e{-7}$	$4.54770919e{-3}$

As shown in Table I, when the time step is very large ($h = 0.5$), there is a significant discrepancy of 30.45% between the simulation and the theory. This noticeable error happens because our theoretical formula is derived using lowest order perturbation theory, which assumes that the step size h is extremely close to zero. When a large value like $h = 0.5$ is used, higher order truncation terms become strong enough to influence the system. These neglected mathematical terms alter the true behavior of the numerical orbit, causing the simulated precession to be slower than the theoretical prediction. However, as h is reduced to 0.2 and 0.1, the relative error drops rapidly to 5.72% and 1.78% respectively, showing that these higher order effects fade quickly as the grid becomes finer.

IV. Conclusion

This report demonstrated the numerical reliability and geometric properties of the Leapfrog integration scheme when applied to the Kepler two body problem. Through simulations spanning 100 orbital periods, the structural benefits of symplectic algorithms over traditional numerical methods were clearly validated. Specifically, the standard energy H displayed bounded, repeating oscillations with absolutely no long term artificial drift, which is the defining characteristic of a symplectic integrator. Furthermore, by incorporating a second-order analytical correction term, the modified shadow energy $\tilde{H}$ successfully canceled out high frequency variations. This dramatic reduction in oscillation amplitude provided direct proof of backward error analysis, confirming that the discrete steps follow a highly accurate, altered path described by a shadow Hamiltonian. Simultaneously, the system preserved angular momentum near to machine precision ($\sim\ 10^{-14}$), owing to the exact geometric alignment of the discrete acceleration vector toward the central origin, which completely eliminates numerical torque. The mathematical design of the algorithm was further verified by a formal convergence test. Both the maximum relative energy drift and the peak - to - peak bound scaled quadratically ($\mathcal{O}(h^2)$) with the step size h, confirming that the time reversible structure of the method effectively cancels out local truncation errors. When comparing algorithmic sequences, the alternative ordering $\hat{O}'$ showed a clear performance advantage over the standard $\hat{O}$, reducing the maximum error envelope by approximately a factor of three due to its phase sensitivity at the periapsis starting point. Finally, the investigation of the artificial orbital rotation confirmed that numerical precession follows a systematic $\mathcal{O}(h^2)$ power law. Although larger step sizes revealed visible discrepancies because higher order terms become strong enough to influence the system, smaller step sizes matched our analytical perturbation model with exceptional accuracy, showing errors below 0.12% for $h = 0.01$. In conclusion, these findings underscore the profound importance of preserving geometric invariants when simulating dynamical systems. While high order non-symplectic methods often prioritize short term precision at the expense of long term qualitative physics, the second-order Leapfrog method guarantees structural stability over long timescales.

Acknowledgment

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